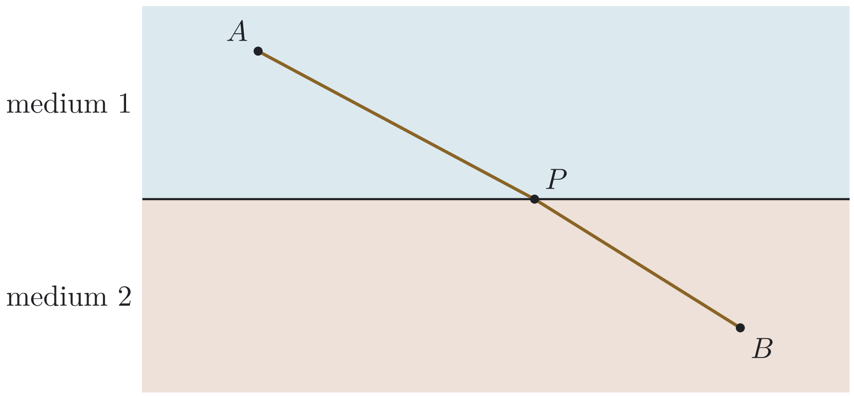
# The Fastest Crossing

Name: ____________________

*Two transparent materials, one boundary, and a law withheld until distance and time have built the case for it.*



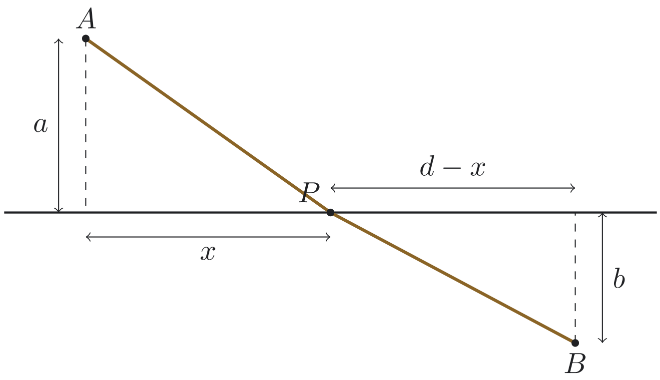

**The question.** Light travels from $A$ in one transparent material to $B$ in another. The two materials transmit light at different speeds. Where should the path cross the boundary if the journey takes the least possible time?

**Deliberately withheld at the start:** the normal, angle of incidence, Snell's law, and refractive index. Your task is to make these ideas necessary rather than receive them as vocabulary.

## 1. Build the time model

distance over speed

Let $a$ and $b$ be the perpendicular distances of $A$ and $B$ from the boundary. Let their horizontal separation be $d$, and let the crossing point be a horizontal distance $x$ from the point directly below $A$.




Use Pythagoras to write each path length:

$$AP = \underline{\hspace{4cm}}, \qquad PB = \underline{\hspace{4cm}}.$$

Hence construct the total travel time:

$$T(x) = \frac{\underline{\hspace{4cm}}}{v_1} + \frac{\underline{\hspace{4cm}}}{v_2}.$$

## 2. Search before differentiating

try crossing points

Use $a = 3.0\,\text{cm}$, $b = 2.0\,\text{cm}$, $d = 8.0\,\text{cm}$, $v_1 = 4.0$ units and $v_2 = 2.0$ units. Calculate the total time for several values of $x$.

| $x$ | $AP$ | $PB$ | $T$ |
|---|---|---|---|
| 0 | | | |
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |

Best crossing point from this table: ____________________

What pattern do you see on either side of the best value?

**3. Change the medium** what moves, and why? Predict what happens to the best crossing point if medium 2 becomes progressively slower while all distances remain fixed.

Explain the prediction in words before doing another calculation.

Complete the qualitative table.

| Medium 2 relative to medium 1 | Predicted path behaviour |
|---|---|
| Same speed | |
| Slower | |
| Faster | |
| Very much slower | |

**4. Invent a medium property** slowness as time cost per metre Define the slowness of a medium by

$$s = \frac{1}{v}.$$

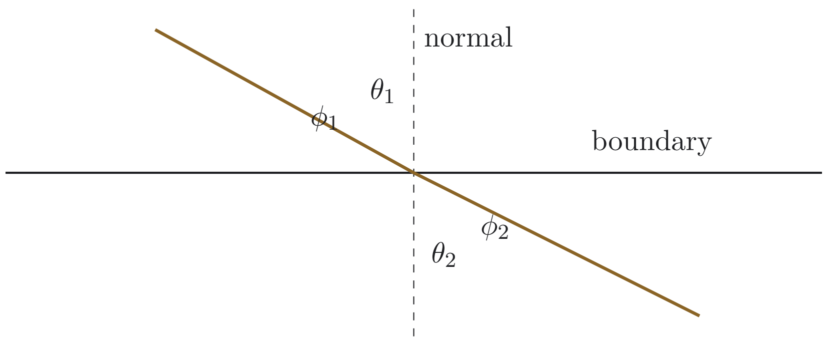
State its units and interpret it physically.

Rewrite the time function using $s_1$ and $s_2$:

$$T(x) = \underline{\hspace{10cm}}.$$

Why might slowness be a more natural quantity than speed in a minimum-time problem?

**5. Decide how angles will be measured** boundary or normal? At the crossing point, draw both possible angle conventions.




Give two reasons for choosing angles from the normal rather than from the boundary.

1. ______________________________

2. ______________________________

## 6. Find the minimum condition

differentiate the time model

Starting from

$$T(x) = s_1\sqrt{a^2+x^2} + s_2\sqrt{b^2+(d-x)^2},$$

find $\dfrac{dT}{dx}$.

At the minimum, set $dT/dx = 0$ and rearrange to obtain

$$s_1 \frac{x}{\sqrt{a^2+x^2}} = s_2 \frac{d-x}{\sqrt{b^2+(d-x)^2}}.$$

Show where the two fractions appear in the geometry and complete:

$$\frac{x}{\sqrt{a^2+x^2}} = \_\_\_\_\_\_\_\_, \qquad \frac{d-x}{\sqrt{b^2+(d-x)^2}} = \_\_\_\_\_\_\_\_.$$

Hence write the minimum-time condition in two equivalent forms:

$\boxed{\qquad\qquad}$ and $\boxed{\qquad\qquad}$.

## 7. Interpret the law

the formula must agree with the model

Use your minimum-time condition to explain each case.

$v_1 = v_2$

$v_2 < v_1$

$v_2 > v_1$

$v_2 \to 0$

Does the law require the familiar symbol $n$? Explain.

## 8. Normalised slowness

where refractive index comes from

The

refractive index is

$$n = \frac{c}{v} = c\,s.$$

Explain why this is a normalised slowness rather than a new physical mechanism.

Show that the minimum-time condition becomes the familiar Snell form.

**9. Extend the model** from two media to a varying medium For many short path elements $\Delta\ell_i$ with local slowness $s_i$, write a total time cost:

$$T \approx \sum_i \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ .$$

In the continuous limit:

$$T = \int \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ .$$

Explain the analogy with walking across terrain in which each metre has a different time cost.

How does this connect back to the fastest-fall problem, where speed changes continuously with depth?

**10. Productive stuck** what did you establish before the final law? Complete at least four entries.

| | |
|---|---|
| **I built a model** | The total time depends on... |
| **I found evidence** | My table or graph showed... |
| **I invented a useful quantity** | Slowness helped because... |
| **I motivated a convention** | Measuring angle from the normal is useful because... |
| **I tested a limiting case** | When one speed becomes very large or very small... |
| **I connected representations** | The geometric fractions became trigonometric functions when... |
| **I identified an assumption** | The derivation assumes... |

## 11. Field notes

talk generates; writing stabilises

Before discussion, write:

I first thought the shortest path would be...

The first result that challenged or refined this was...

The quantity I found most useful was...

After discussion, write:

Another person's argument changed my model because...

My current explanation of refraction is...

The step I can justify least securely is...

## 12. Core results what this enquiry actually earned

**The relations established along the way.**

$$s = \frac{1}{v} \qquad T(x) = s_1\sqrt{a^2+x^2} + s_2\sqrt{b^2+(d-x)^2}$$

$$\boxed{s_1 \sin\theta_1 = s_2 \sin\theta_2} \iff \boxed{\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}} \qquad n = \frac{c}{v}$$

**Concepts earned, in order.**

- A minimum-time argument, not a reflection law, is what actually decides where light crosses a boundary.
- Slowness, the reciprocal of speed, turns the time function into a simple weighted sum of distances.
- The normal is a motivated choice of angle, not an arbitrary convention: it is the one that makes a straight-through ray read as zero degrees.
- Refractive index is a normalised slowness, $n = c\,s$, not a separate physical mechanism.
- The same reasoning, applied to a continuously varying medium, is exactly what the fastest-fall problem already needed.

**Every substantial claim should be accompanied by a reason, calculation, diagram, table, limiting case, or counterexample.**